



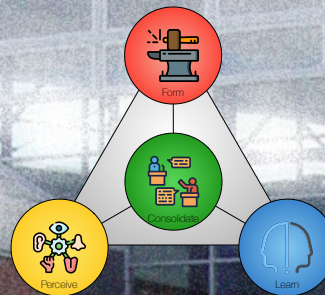
AALBORG UNIVERSITY
DENMARK

11 MERIT

ESSENCE – CHAPTER 18

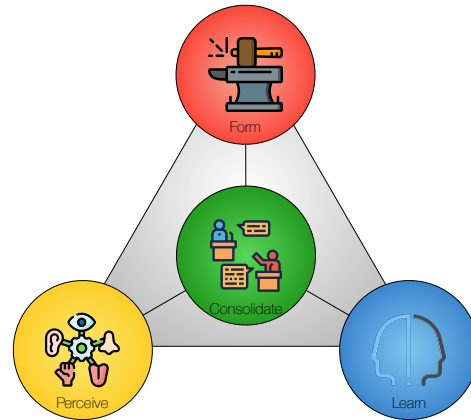
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SWI 2026



Agenda

- Exercise 10 Follow-up.
- Chapter 18: Merit.
 - Merit Concepts (functional part of PSC).
 - Robustness (non-functional supplement).
 - Dependencies (non-functional supplement).
 - The ETVX Model (non-functional supplement).
- Next time:
 - Exercise 11: Merit.
 - Lecture 12: Innovation as Design Science.

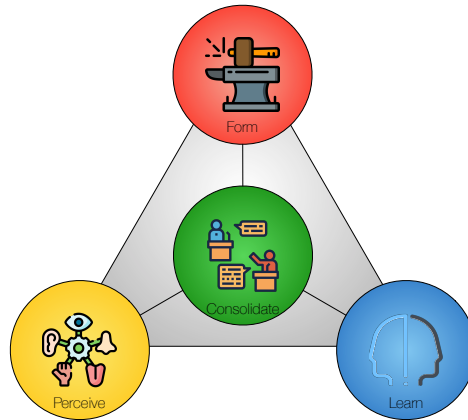


Follow-up: Exercise 10

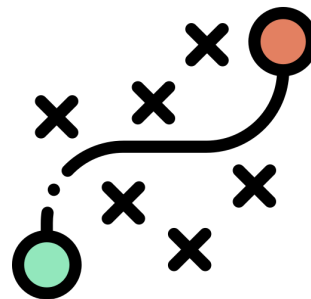
Exercise 10 (Tactics, Manifestations & Capabilities)

This exercise concerns the **tactics**, **manifestations**, and **capabilities** of your project.

- Will you characterize your approach as **push** or **pull**?
- Describe concrete and abstract **manifestations**.
- Describe concrete and abstract **capabilities**.
- Will your abstractions improve **evolvability** and make your design **future-proof**? How?



Tactics



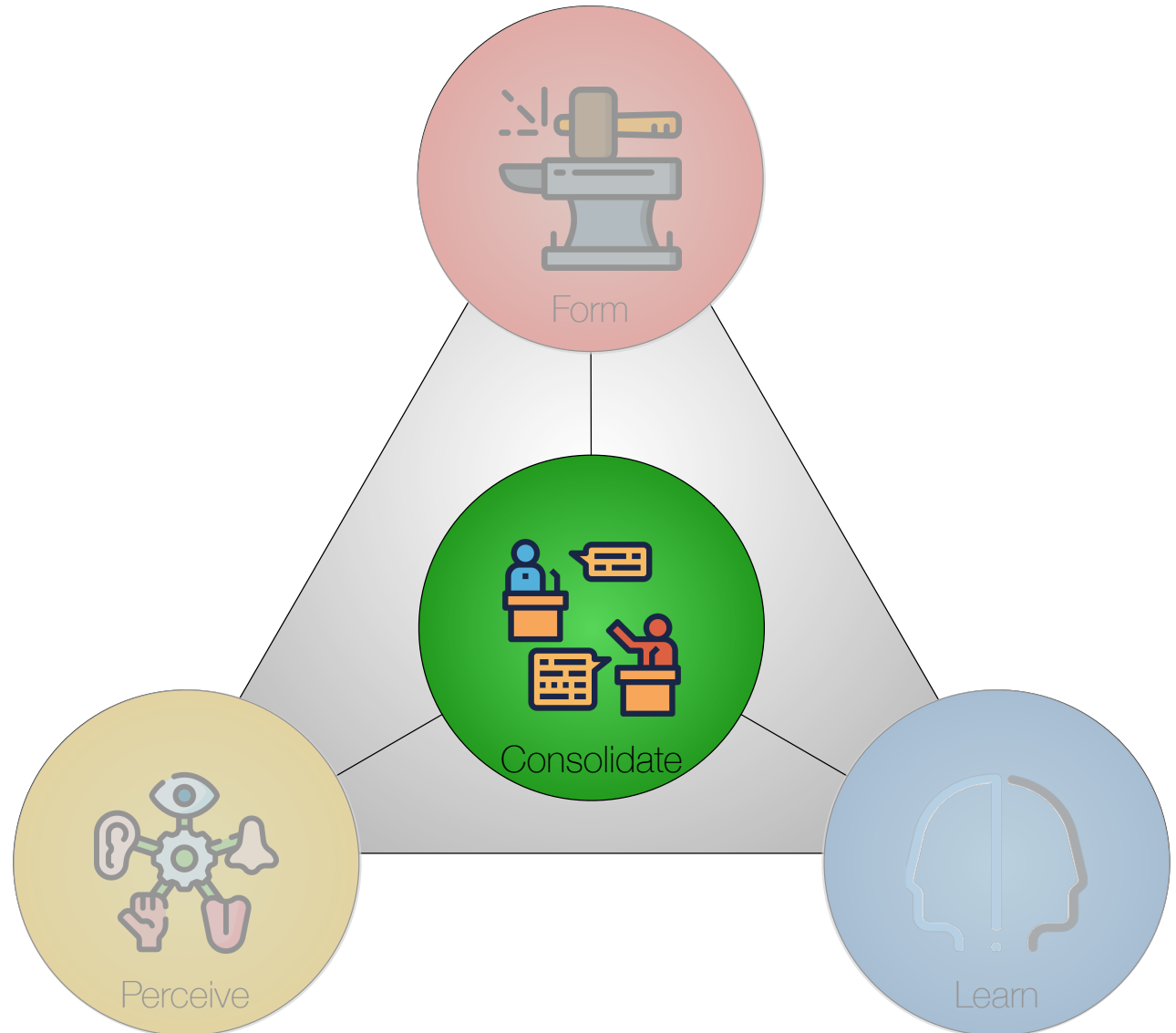
Do the difficult things while they are easy and do the great things while they are small.

A journey of a thousand miles must begin with a single step

Lao Tzu (6th century BC–5th century BC)

Four Core Activities

- *Perceive* – understanding the problem and its context.
- *Form* – designing the constructive parts of a solution.
- *Consolidate* – integrating the design contributions into a clear whole that addresses the situation.
- *Learn* – appraising the design.



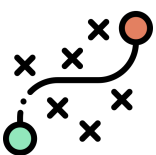
Three Levels of Abstraction



Rationale – the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the overall problem.

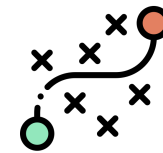


Strategy – the **overall plan** for addressing the problem, including its **scope** and **context**, the essential parts needed to build the solution, and any limitations or constraints that might influence its feasibility. At the strategy level, we also evaluate the potential for future **growth**.



Tactics – actions planned in accordance with the strategy to achieve specific ends.

Consolidate Terminology

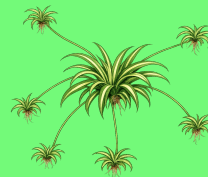


Solution



Why the problem is resolved, why it is important, and why the result is appealing.

Evolvability

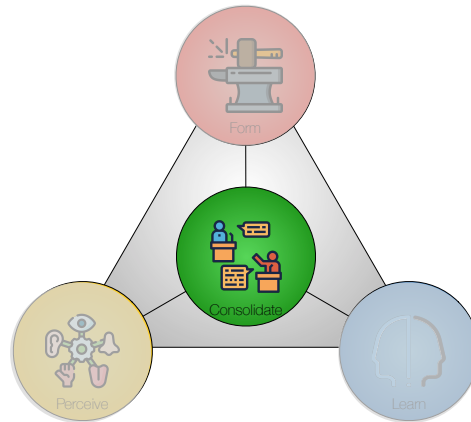


The capacity to adapt to outer environments or to add new leverage points to the inner environment.

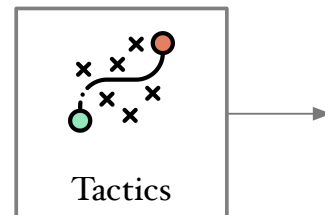
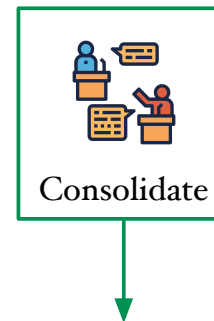
Merit



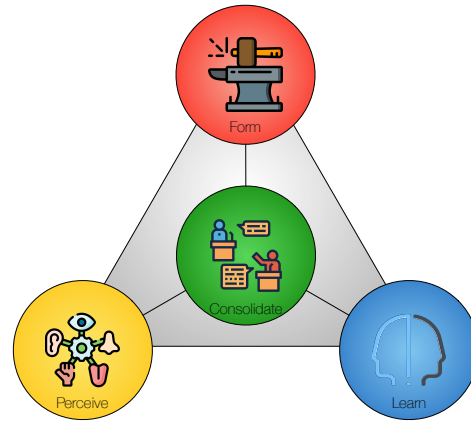
The value offered by the capabilities. The limitations of the solution and why these limitations are acceptable.



Merit



*Better a diamond with a flaw
than a pebble without*
Confucius (551BC–479BC)



Merit Concepts

Functional and Non-functional Requirements

- Functional requirements are typically first-order concerns (part of PSC):
 - Is the solution functionally complete? Can we accept limitations?
- Non-functional requirements are typically second-order concerns (supplement to PSC):
 - Will it operate effectively?
 - Is it vulnerable to significant risks?

Stakeholders usually consider function primary—what they pay for.

Functional Merits

Functional requirements: Is the solution functionally complete? Can we accept limitations?

- **Value**—the functional value of the capabilities.
- **Qualifications**—a qualification is a statement or assertion that makes another less absolute.
- **Reservation**—limitations in the solution (why the design might not support a complete solution to the problem).
- **Rebuttal**—why limitations to the solution would be acceptable anyway.

Functional Fit

Qualification is the **degree of fit** between the inner and outer environment.

- Does the functionality match what the **situation** requires?
- Are there **reservations** about this fit?
- Can we, as a **rebuttal**, state why we could accept these?

Functional Fit and Tactics

- Does the **manifestations** cell cover the essential parts of the problem?
- Does the **capabilities** cell cover the essential tools or features required to handle these manifestations?
- Does the **value** explain what value the manifestations and capabilities provide?

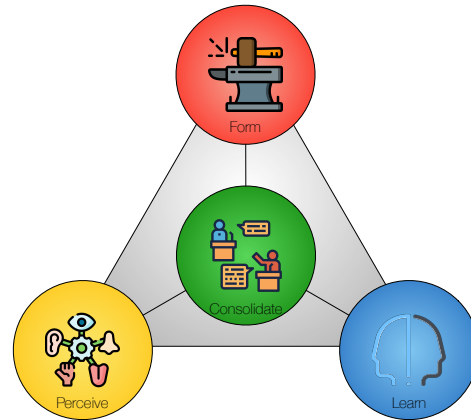
Formal relations

Manifestations, Capabilities, Values

- Every **manifestation** is linked to at least one capability.
- Every **capability** is linked to at least one manifestation.
- Every **capability** is associated with at least one value.
- Every **value** is associated with at least one capability.

Non-functional Merits

- Non-functional features may help prepare for unforeseen situations.
- Non-functional features may also introduce vulnerabilities and dependencies into our design.
- **Robustness**—what enables a system to withstand adverse or unexpected conditions.
- **Dependencies**—vulnerabilities caused by factors outside our immediate control.



Robustness

What enables a system to withstand adverse or unexpected conditions.

Robustness Defined

- Robustness means **insensitivity** to damages or the **ability to withstand** adverse and unforeseen events and consequences of human error without damage (SEBoK, 2024).
- In Essence, we will slightly expand this definition to encompass the **solution** we deliver and our development **strategies**.
- We do this to **prevent or mitigate risks** and to enhance the **capacity to adapt** to future technological changes, available services, working conditions, and more.

Architectural Strategies

- **Near-decomposability** to reduce the chances of failures in one module cascading into others, reduce the technical debt, and simplify future changes.
- **Graceful degradation** so that functionality is maintained to a meaningful degree when parts of the system fail.
- **Cognitive load management** to avoid overloading users and thereby reduce mistakes.
- **Automation with user control** that allows users to override or adjust system outputs when errors arise.
- **Error prevention and recovery** with undo options, informing users through error messages, and guided corrections.

Operational Strategies

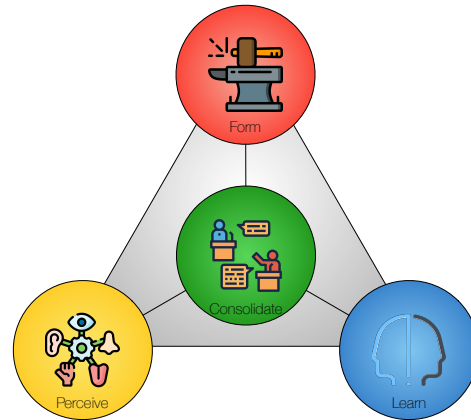
- Continuous monitoring to detect anomalies, address performance issues, and more.
- Also helpful in predictive maintenance, performance optimization, and risk management and mitigation.
- Regular stress testing under realistic conditions to assess the robustness of critical systems.
- Designing alternative pathways for critical processes to enhance robustness and resilience.
- This includes creating backup plans or alternative routes to be activated in case of disruptions.

Adaptive Strategies

- **Self-healing features** to strengthen the system's resilience through fault detection and automated responses.
- **Predictive analytics** to anticipate and mitigate failures by forecasting trends and potential disruptions.
- **Adaptive user interfaces** are concerned with creating interfaces that adjust to user experience level, environment, fatigue, urgency, stress, and more.

Adaptive Strategies 2

- **Multimodal interaction** using speech, gestures, keyboard, and more to ensure accessibility and allow users to select their preferred modes of interaction.
- **Context awareness** making the system adjust to emergencies in health care, critical infrastructure, or cockpits.
- **Role-based interfaces** providing relevant data to users, such as doctors, nurses, and technicians.



Dependencies

Vulnerabilities caused by factors outside
our immediate control

Dependencies

- Dependencies are **vulnerabilities** that arise from factors beyond our immediate control.
- These factors may stem from both the **outer** and **inner** environments.
- **External services** or **internal leverage points** on which our design depends are potential vulnerabilities:
 - When they work, they may help us build stronger solutions than we could on our own (for example, security services).
 - Ironically, potential **disruptions** of such could render our solutions **crippled** or **useless**.

Risk Assessment

- **Likelihood assessments** analyze general trends and forces.
 - **Trends** in technology, regulations, and society, as well as **forces** within industries, markets, and general economics, that could influence the **availability** or **vulnerability** of services, platforms, components, and more.
- **Impact assessments** analyze the direct and indirect effects of vulnerabilities.
 - **Direct** effects on core functionalities could cause failures or degraded functionality, service interruptions, data inconsistencies, etc.
 - **Indirect** effects of dependency could harm revenues, customer satisfaction, or requirements compliance.

Mitigation Strategies

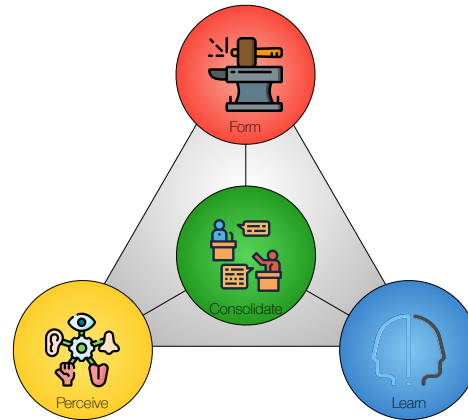
- Designing for **graceful degradation** when a dependency fails.
 - Circuit breakers, retries, and such when they fail.
 - Fallbacks that maintain core functionalities and reduce the likelihood of failing.
- A modular architecture based on **near-decomposition** to limit the propagation of failures and minimize the impact.
- Adjusting **system boundaries**:
 - Modifying the scope to **exclude** a risky feature or to **include** a reliable internal solution rather than relying on a high-risk external service.

Managing External Dependencies

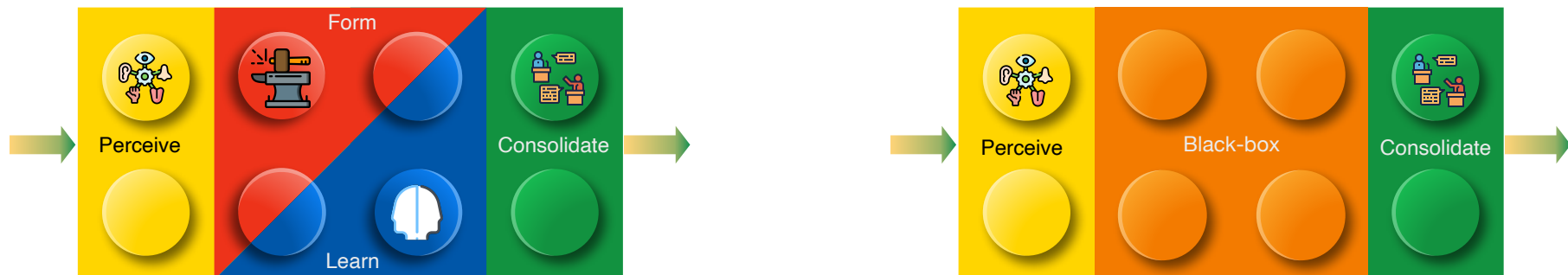
- Establish clear integration boundaries using **abstraction layers** to protect from external variability.
- Ensure external services are used following **well-defined standards**.
- Shield the architecture from external changes and make interactions explicit and manageable. Use **boundary definitions**.
- Apply design strategies for **graceful degradation** and **fallback**.

Managing Internal Dependencies

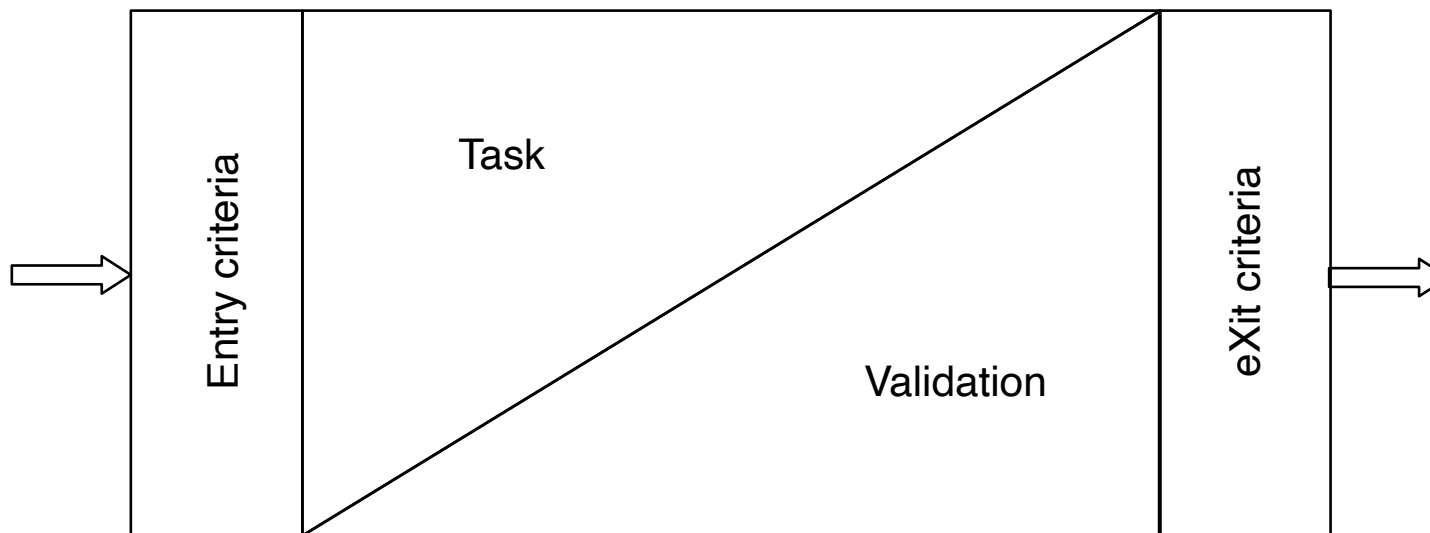
- Map, assess, and mitigate **leverage point** dependencies.
- Emphasize **modularity**, encapsulate functionalities, and map dependencies to ensure transparency, maintainability, scalability, and resilience.
- Use **separation of concerns** to make dependencies explicit and easier to manage.
- Use **abstractions** extensively and focus on core functionalities and principles. Build strong mental models to manage complexities.
- Apply **information-hiding** principles to conceal internal details and expose only the necessary interfaces.



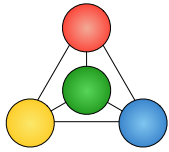
The ETVX Model



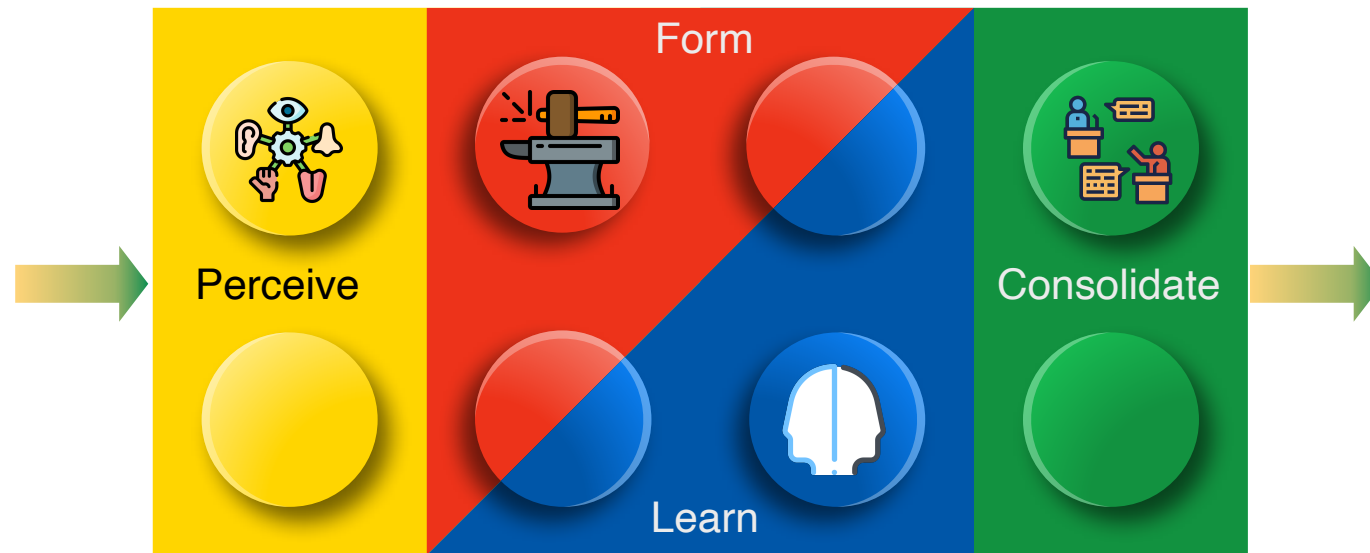
General (ETVX) model



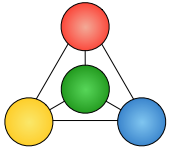
- **Entry** Criteria – criteria that should be satisfied before beginning the task.
- **Task** – descriptions of what is to be accomplished.
- **Validation** – procedures to ensure the quality of what is accomplished by the task.
- **eXit** – criteria that should be satisfied for the activity to terminate.



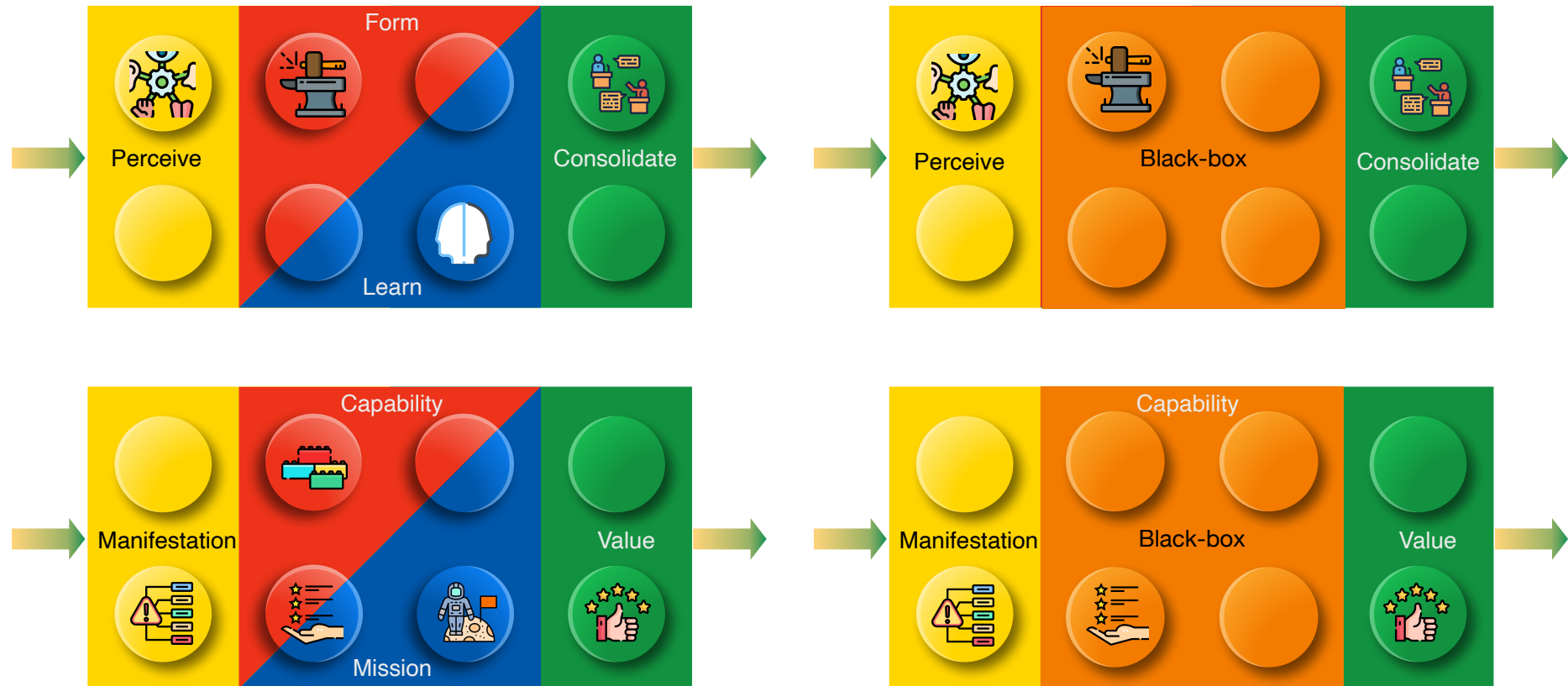
Essence adaptation of the ETVX model



- **Entry** Criteria – criteria that should be satisfied before beginning the task.
- **Task** – descriptions of what is to be accomplished.
- **Validation** – procedures to ensure the quality of what is accomplished by the task.
- **eXit** – criteria that should be satisfied for the activity to terminate.



Model Variants

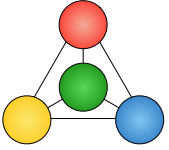


White-box cells: where the details of the contribution and valuation are known.

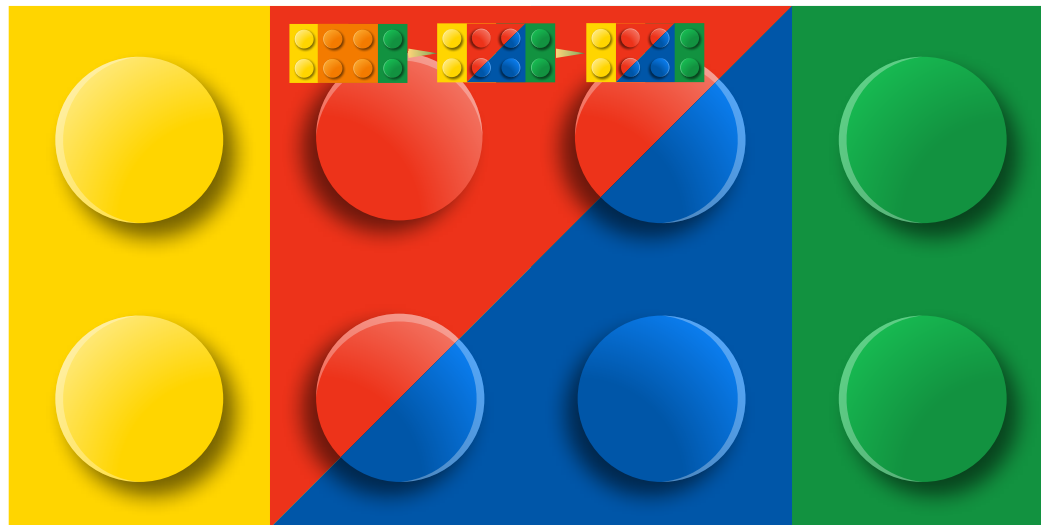
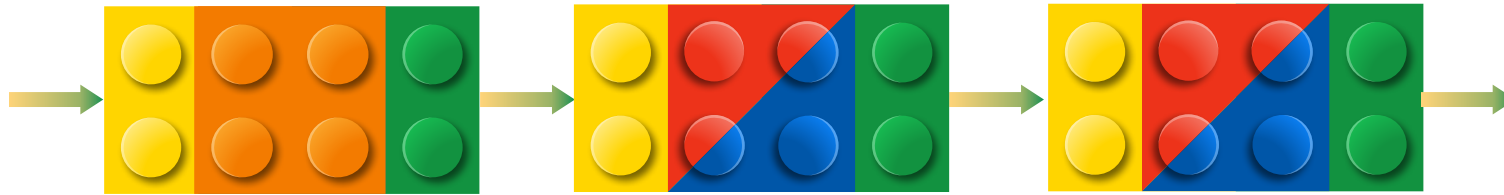
Black-box cells: where we will not know these details.

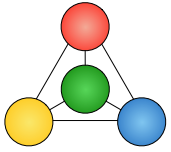
Black-box examples:

Third-party components or services, outsourced sub-projects, external customer organizations.

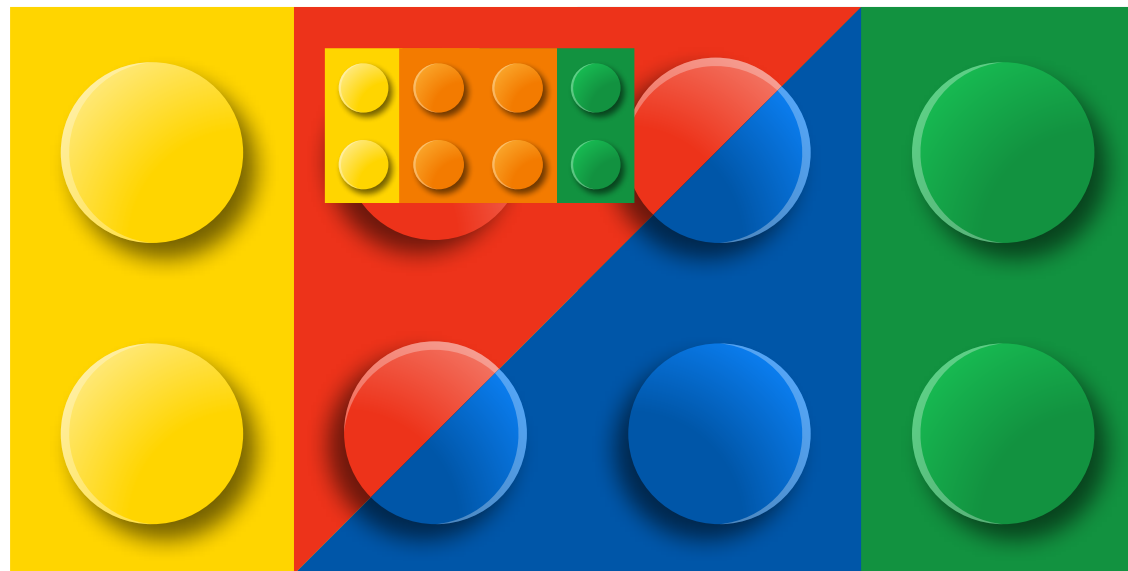


Cascading Cells

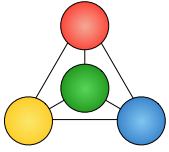




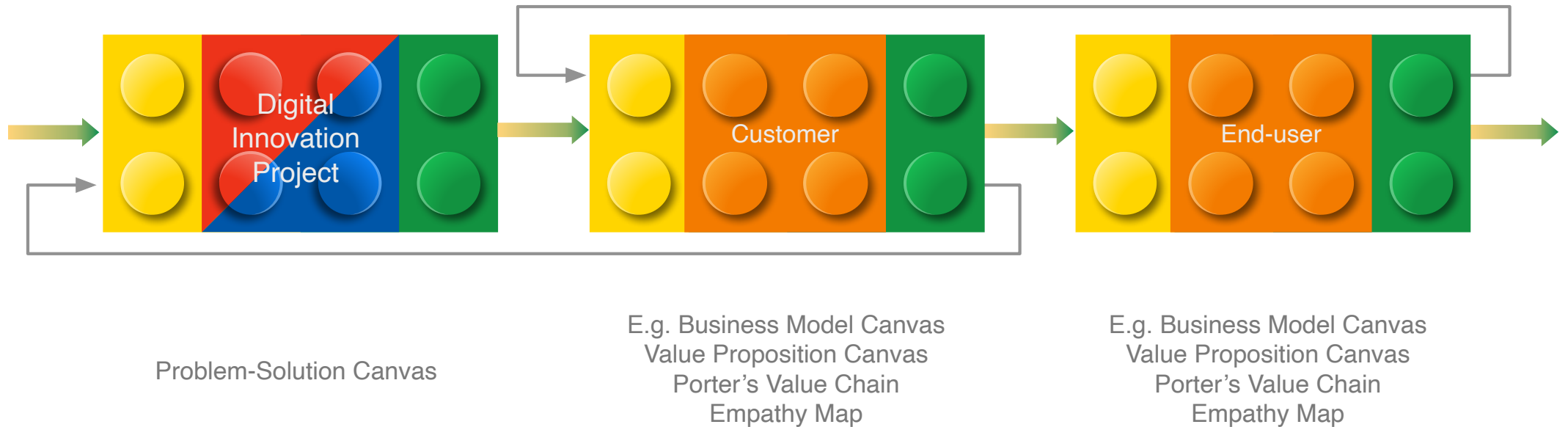
Abstraction Layers



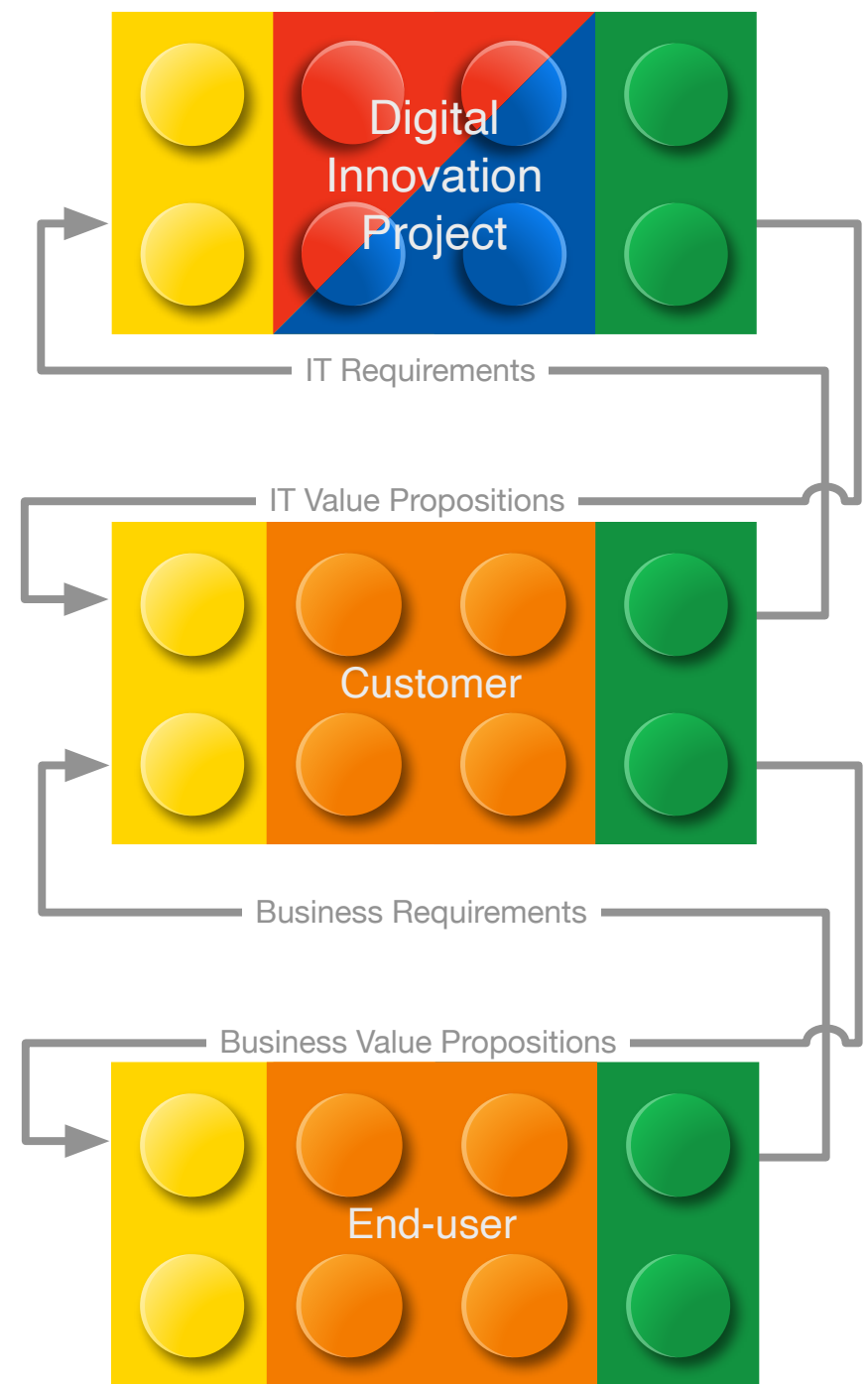
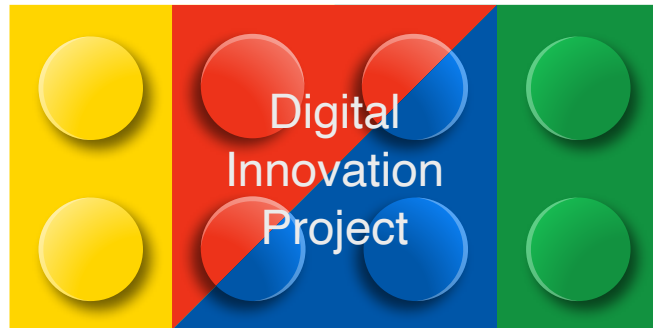
Abstraction layers **conceal** the operational details of external components and allow for **replacing** one external element with another or an internal component.



Communicating vessels



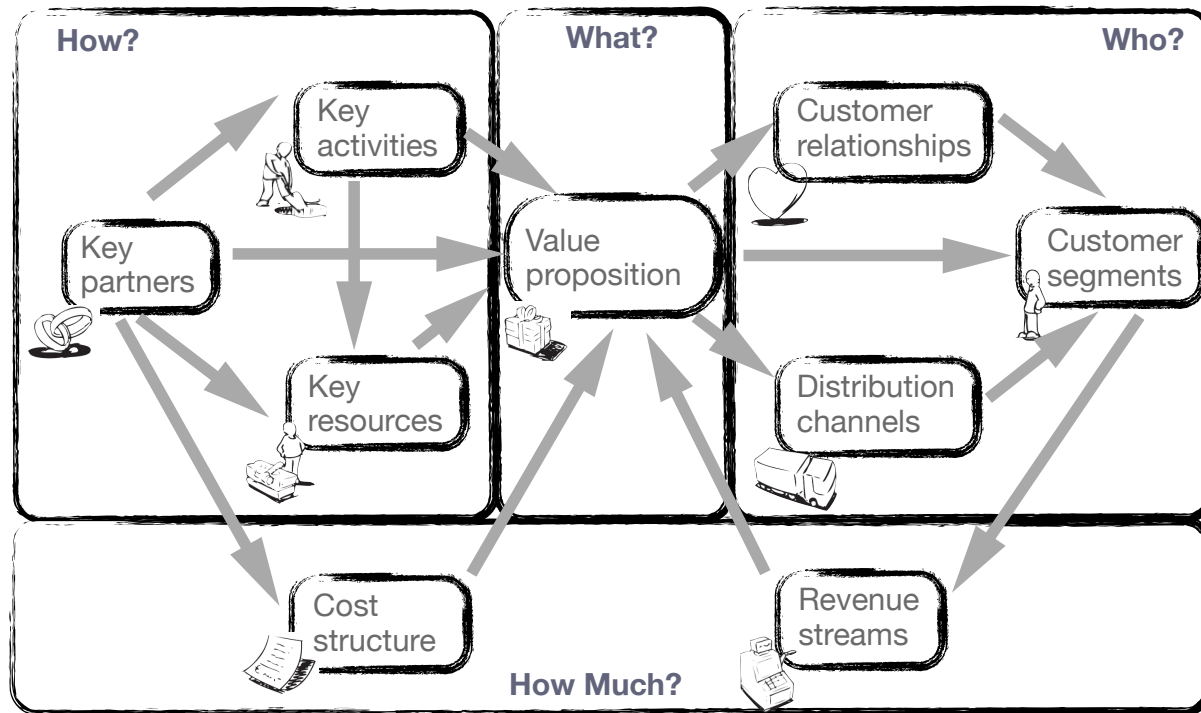
Reasons to Pivot



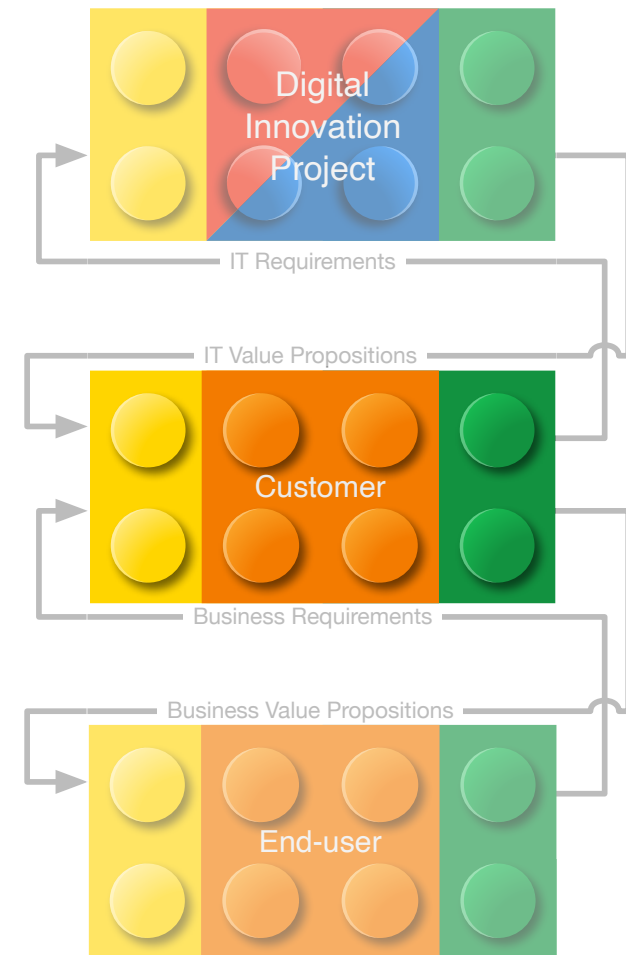
**Internal
inconsistencies**

**External
dynamics**

Business Model Example



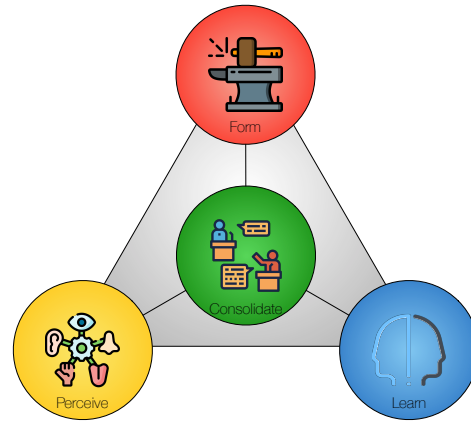
www.businessmodelgeneration.com



Changes in any of the nine elements of this model might cause us to change the Problem-Solution Canvas

Essence Pivots

- *Zoom-in pivot*, where the contribution is narrowed to include fewer features.
- *Zoom-out pivot*, where the contribution is widened to include more features.
- *Leverage pivot* using new or exploiting better existing leverage points (technologies, components, information, human resources).
- *Outer Environment pivot*, where the boundaries of the situation are altered concerning outer services, implements, repositories, or people and practices.
- *Platform pivot*, where the inner environment is opened for third parties, such as external partners, to build the contribution as a platform for expanding ecologies.
- *Manifestation pivot*, where customers/users are unchanged, although the project now focuses on other needs than first expected.



Next Time

Next Exercise and Lecture

- Exercise 11:
 - Merit.
- Lecture 12:
 - Innovation as Design Science:
 - Hevner, A. & Gregor, S. (2022). Envisioning entrepreneurship and digital innovation through a design science research lens: A matrix approach. *Information & Management*, 59(3), 103350.
 - Course overview.

Exercise 11 (Merit)

This exercise concerns **functional** and **non-functional** requirements, as well as strategies for improving **robustness** and managing **dependencies**.

- Discuss the **functional** completeness of your design and state **reservations**, **rebuttals**, and **values** accordingly.
- Discuss the **robustness** of your design and assess if any of the strategies in section 18.2 would be helpful.
- Outline **dependencies** in your design and assess the utility of the strategies in section 18.3.
- Use **ETVX** (section 18.4) to model **robustness** and **dependencies**.